A Hybrid Multi-Stage Machine Learning System for Accessible Thyroid Pre-Screening

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***Abstract*—Among the most prevalent endocrine diseases are thyroid disorders. Hence, there is a need to develop effective diagnostic tools for the management of this disease. This study presents a machine learning–based approach for developing a thyroid disease screening system. The system incorporates a deep learning model to perform classification of medical images. In addition to this, a weighted symptom scoring mechanism has also been integrated into the system for the evaluation of symptoms. In this system, EfficientNetB0 [2] has been employed for the classification of images. In addition to this, a weighted symptom scoring mechanism has also been integrated into the system for the evaluation of symptoms. In this system, a weighted scoring mechanism has been employed for the evaluation of symptoms. In addition to this, a Random Forest classifier [11] has also been integrated into the system for the evaluation of biochemical hormone level data. Within this framework, the hormone-based model achieved the highest accuracy of around 83%. When all components of the system are considered together, the overall accuracy is approximately 77%. The results show the balance between diagnostic accuracy and accessibility. The proposed system offers a practical and scalable approach for early thyroid screening, particularly in healthcare environments where resources are limited.**

***Index Terms*—Thyroid Detection, ML, DL, Random Forest, EfficientNetB0 CNN, Throat Images, Pre-screening.**

I. Introduction

Thyroid diseases are among the most common endocrine diseases and affect millions of people around the world. If thyroid illnesses aren't detected soon enough, the body's metabolic and physiological functions can be seriously harmed. The thyroid gland is an essential organ that regulates metabolic activity and growth through the production of its main hormones, triiodothyronine (T3) and thyroxine (T4). When the levels of thyroid hormones are altered due to a hormonal imbalance, patients will subsequently develop conditions like hypothyroidism and hyperthyroidism and thyroid nodules that require medical attention or they may suffer serious medical complications. The diagnosis and treatment of thyroid conditions requires the identification of thyroid function problems as well as ongoing monitoring of those problems during the patient's treatment course.

Standard diagnostic methods for determining the presence of thyroid disorders exclusively involve laboratory hormone testing and ultrasound imaging, as these two methods provide reliable information. However, both methods rely on specialized diagnostic machines, trained technicians, and specific health care environments; therefore, patients living in rural and/or economically challenged areas would have difficulty accessing these resources to evaluate and diagnose thyroid dysfunction. These individuals need improved access to less expensive testing methods capable of performing basic evaluations prior to clinically validated testing.

Artificial intelligence (AI) and machine learning (ML) have improved medical diagnostic systems in recent years. They do this by finding Hidden Patterns from Complex Data Sets that Regular Techniques Won't Find. Some Deep Learning Techniques - such as Convolutional Neural Networks (CNN's) can create a Classification of an Illness and an Identification of the Presence of any Illnesses in Images that Were Originally Generated As An Image in the First Place (See References 1 and 2). Random Forests and Support Vector Machines are commonly used algorithms for Structured Clinical Data as these algorithms can be Used To Model Non-Linear Relationships and to Model Variations in Medical Data (See References 3 and 6).

Thyroid Prediction Systems Continue To Be Based On Biochemical Hormone Data As The Required Testing At A Laboratory Creates A Barrier To Early Screening. Biochemical Hormone-Based Models Have Shown To Have High Accuracy, Due to The Data Used Until Now Has Been Structured And Validated. Image And Symptom Based Systems Have Shown Greater Accessibility; However, Their Accuracy Of Prediction Is Reduced Due To Data Quality And The Restricted Nature Of Data Access.

This article describes a hybrid ML approach that leverages many types of data to provide screening for thyroid disorders based on multiple sources of relevant information. The hybrid approach includes a deep neural network called EfficientNetB0 to evaluate neck images along with a weighted questionnaire module which assesses symptoms, and a Random Forest classifier which determines the expected Time-of-Day (TOD) for the measurement of hormone data.

The proposed methodology combines multiple input types into an iterative ('looped') workflow. Initially, all screening activity will take place using the available visual and symptom-based input. Only after these screening activities are completed will the system prompt the user for hormone data.

The hybrid system described in this paper accomplishes its goals for accurate diagnosis and timely assessment by combining unique processes that greatly enhance accessibility to real-world environments.

The remaining sections of this paper are organized as follows:

In Section II, we will review previously published studies relative to both the prediction of thyroid disease as well as those studies on various types of ML applications within healthcare. We will describe our overall methodology in Section III, and system architecture and implementation in Section IV. Section V provides the experimental results as well as discussion of both how the system performed and future performance evaluations. Finally, Section VI contains conclusions from this study and ideas for future studies.

II. Related Work

The integration of Artificial Intelligence (AI) and Machine Learning (ML) technologies has enabled a revolutionary improvement to the prediction of diseases and diagnosis of patients within the field of medicine. The research explored the use of ML approaches to develop a classification system for Thyroid Disease(s) utilizing structured and clinical datasets. In many cases, traditional methodologies of biochemical hormone testing used to measure levels of Thyroid Stimulating Hormone (TSH) (which provides an indication of thyroid function), triiodothyronine (T3) and Thyroxine (T4) are used by health professionals to diagnose patients with Thyroid Disease(s). SVM (Support Vector Machines) as well as Decision Trees and Random Forest Classifiers were utilized in this research to assess the dataset utilizing the common ML methodologies. According to sources 3, 4 and 11, the proposed system is not limited to assessing linear relationships, and thus, improves upon the overall accuracy of classification of patients.

In recent years, medical image analysis has increasingly incorporated deep learning techniques alongside traditional machine learning methods. Applications in areas such as radiology and disease classification have demonstrated the effectiveness of Convolutional Neural Networks (CNNs) [2],[9] perform well in identifying anomalies from medical images [1], [2]. The majority of image-based research on thyroid disorders focuses on ultrasound imaging, which offers intricate internal views of the thyroid gland but necessitates specialized tools and qualified personnel.

There has been a rise in recent studies examining hybrid methods that utilize different data types (e.g., clinical vs imaging) to assist in improving the predictive accuracy of a diagnosis. By utilizing complementary sources of data from different modal types, multimodal systems seek to improve the diagnostic process and decrease classification uncertainty [5][6],[14]. The majority of these hybrid multimodal approaches generally utilize structured clinical/laboratory data; therefore, they may be less applicable for early-stage screening or in resource-limited settings.

While there are many advances being made in developing accessible thyroid screening technologies that combine non-invasive image-based analysis and symptom-based assessment with conditional incorporation of hormone data to provide a more reliable evaluation, there continues to be a gap related to the development of such systems. Current screening systems either provide high-accuracy hormone-based predictions or utilize specialized imaging techniques, both of which may not be appropriate for use in either a large-scale or developmental screening environment.

To mitigate this gap and provide a possible solution to this significant barrier, we propose an innovative hybrid framework that utilizes image-based detection, symptom-based assessment, and conditional hormone-based classification together. This approach provides an opportunity to balance accessibility and diagnostic accuracy for use in all real-world screening modalities where laboratory tests cannot be readily performed [13].

III. SYSTEM ARCHITECTURE AND IMPLEMENTATION

Here's how the team built the hybrid Machine Learning and Deep Learning system to spot thyroid disorders. They set it up as a multi-stage process, blending image analysis, symptom checks, and hormone-level reviews. This combination doesn't just boost the accuracy of diagnoses—it also keeps things straightforward for doctors who need to use it in real clinics.

A. Gathering and collecting data

This pilot study used two main methods to make sure that screening was strong and used more than one method:

1) Set of Images: We pulled together a small set of neck images from public sources, sorted into six groups: Goiter, Hypothyroidism, Thyroid Nodules, Hyperthyroidism, Swollen Neck, and Normal Neck. Including healthy, normal images mattered—it helped balance things out and kept the model from just learning to spot "sick-looking" necks.

2) Data on Symptoms and Hormones: We designed a weighted questionnaire to gather symptom details, focusing on key signs like fatigue or changes in weight. On top of that, for training the next stage of the model, we used TSH, free T3, and free T4 values—either taken from open datasets or simulated if we couldn't find enough real data.

Since big medical image datasets are tough to come by, we jumped in with data augmentation, basically making new, slightly tweaked versions of what we had. This approach helped fill out our dataset and added variety, making it easier to show that the machine learning model could actually do the job.

TABLE I
SUMMARY OF DATASET FEATURES

Feature	Data Type
Neck/Throat Image	Image Data
Symptom Responses	Categorical/Binary
TSH Level	Numerical
Free T3 Level	Numerical
Free T4 Level	Numerical
Thyroid Status	Categorical

B. Data Preprocessing

Preprocessing methods were used on all data types to make them more consistent and improve the performance of the model.

1) Image Preprocessing:

- Image Loading and Resizing: All images came in different formats—like .webp and .jpeg—but we converted them to a standard size, say 224×224 pixels, using bicubic interpolation. This way, important anatomical details stayed sharp.
- Normalization: We scaled the pixel values down to a [0,1] range. Doing this helped the training go smoother and kept things stable.
- Data Augmentation: To make sure the model didn't just memorize the training set, we threw in some data augmentation: rotations, flips, zooms, and tweaks to brightness and contrast. This made the model better at handling new, unseen images.

2) Dataset Splitting: We split the whole dataset into three parts: training got 70%, while validation and test each got 15%. We used stratified sampling for this, so every class was still represented fairly—nobody got left out.

3) Class Balancing: Finally, to avoid favoring majority classes, we calculated class weights and used them during training. This boosted the model's ability to spot conditions that don't show up as often

4) Symptom and Hormone Processing: Questionnaire responses and hormone values were normalized and scaled to maintain consistency across the system.

C. Feature Extraction and Module Development

The system uses both deep learning and heuristic approaches for feature extraction.

1) Image Feature Extraction: A transfer learning approach was adopted using a pretrained EfficientNetB0 [2],[9] convolutional neural network. The base layers were frozen to utilize the pretrained ImageNet features, and a custom classification head was added.

2) Feature Processing Head: The classification head includes a Global Average Pooling (GAP) layer followed by a Dropout layer (50%) to enhance generalization and manage high-level features before final classification.

3) Symptom-Based Feature Representation: The questionnaire module employs a weighted scoring mechanism where each symptom contributes proportionally to an initial risk score based on its clinical significance.

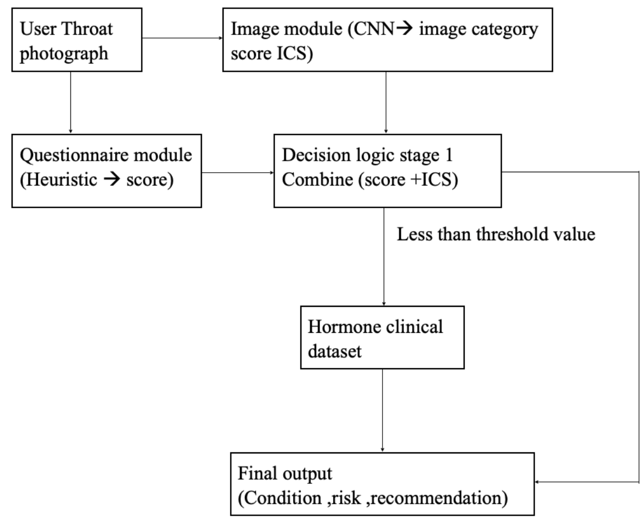

Fig. 1. Data Flow Diagram.

D. Stage 1: Primary Screening (Image + Symptom Fusion)

The first stage of the system conducts preliminary screening by merging outputs from the image and symptom modules.

- A composite risk score (R) is created by combining the image classification probability (ICS) and the symptom-based score.
- The fusion uses a weighted linear combination with a tunable coefficient (alpha) that controls the influence of each modality.
- Before fusion, the softmax outputs of the CNN are calibrated with techniques such as Platt scaling to ensure trustworthy probability estimates.

This stage produces an initial prediction along with a risk classification.

E. Conditional Decision Mechanism

A decision logic layer determines if further analysis is needed.

- When a Stage 1 risk score has "high confidence" it produces a direct output from the system.
- If the Stage 1 score has "intermediate or borderline confidence" the system will then proceed to Stage 2 before producing a final output.
- This conditional flow increases efficiency by limiting laboratory-based requirements for clinically uncertain cases.

F. Stage 2: Hormone-Based Multi-Modal Screening

Some individuals may require additional details via biochemical analysis to determine if they are hypothyroid or hyperthyroid (the second phase) and have additional analyses performed on them.

1) Rule-Based Logic: A set of rules is then used to create the rules-based scoring model based on established endocrinology protocols, which generates a preliminary probability score called R that assesses the likelihood of being hypothyroid or hyperthyroid through laboratory analysis of hormone levels.

2) Machine Learning Model (Random Forest): A Random Forest Classifier [11] was created using a set of hormone values (TSH, T3, T4) that were normalized to create a classifier trained to learn more complex relationships and perform better than using only the preliminary score alone.

G. Final Fusion and Prediction

The final diagnosis is made by integrating the results of both stages.

- The Stage 1 and Stage 2 scores (R1 and R2, respectively) are combined through a weighted decision-level fusion approach.
- When there is biochemical evidence, the fusion will be slightly biased in its favour.
- The output of the fusion will contain: 1) The predicted thyroid condition, 2) The risk level that has been calculated, and 3) Any action that is recommended as a result of the diagnosis, such as referral for further consultation or repeat testing.

H. System Components and Data Flow

The system comprises interconnected modules that are coordinated within a decision logic layer.

- Image module (EfficientNetB0): Processes throat images to provide the likelihood of classification.
- Questionnaire module: Produces an initial risk score based on clinical signs/symptoms.
- Decision logic layer: Manages the (conditional) flow of information between stages.
- Hormones (Random Forest + Rule-Based): Provides refined predictions through the use of biochemical information.

The overall data flow goes sequentially from input acquisition to final decision generation, ensuring both reliability and flexibility.

I. System Workflow Summary

The complete execution flow of the system is as follows:

- Input acquisition (image + symptoms)
- Data preprocessing
- Feature extraction (image + symptom scoring)
- Stage 1 fusion and initial prediction
- Conditional decision check
- Stage 2 hormone-based refinement (if required)
- Final fusion and output generation

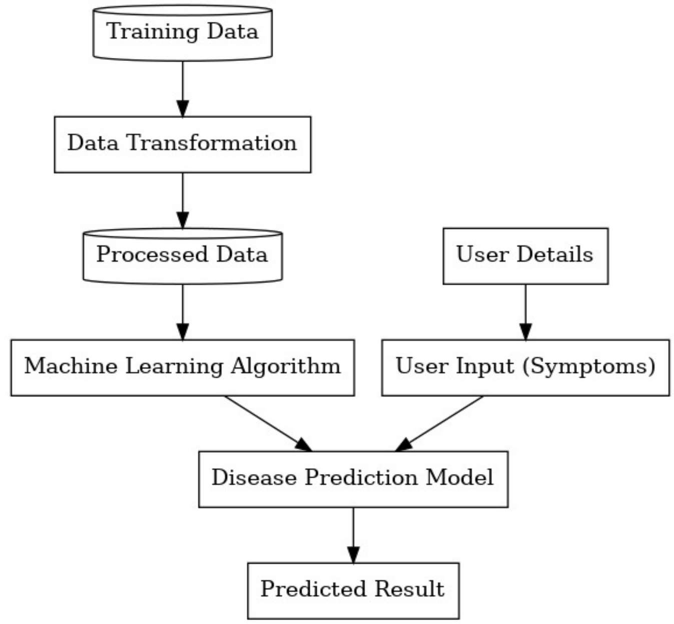

Fig. 2. System Architecture Diagram.

IV. MODEL EVALUATION AND RESULTS

In order to thoroughly assess the effectiveness of the suggested two-step procedure, the parts of DL-based (EfficientNetB0) and ML-based (Random Forest using hormones) were trained, fine-tuned, and compared. The assessment was done based on the usual classification metrics of accuracy, precision, recall, F1-score, and ROC-AUC given the critical need for reliable classification in medical diagnostics [13].

A. Evaluation Process

- DL Model: The EfficientNetB0 model was trained through the use of Adam optimizer and sparse categorical cross-entropy loss. For enhancing model robustness, Early Stopping and learning rate scheduling were used.
- The hormone module uses the Random Forest Algorithm [11] and some simple models, such as Logistic Regressions, for comparison.
- Hyperparameter Tuning: A particular tuning procedure was carried out that included the fine-tuning of the upper layers of EfficientNetB0 as well as the optimization of Random Forest parameters (e.g., number of estimators, max depth).
- Validation Strategy: The model used validation-based checkpointing as its mainstay to retain the finest model version and obtain performance estimates that are not affected by any bias.

B. Model Performance Comparison

An ablation study [12] was conducted to evaluate the contribution of individual modules on 'categorical parts of dataset'.

TABLE II
MODEL PERFORMANCE COMPARISON ACROSS MODALITIES

Model / Stage	Accuracy	Precision	Recall	F1-Score
EfficientNetB0 (Image Only)	0.71	0.70	0.72	0.72
Hybrid Stage 1 (Img + Symp)	**0.75**	**0.76**	**0.75**	**0.73**
SVM (Hormone Data)*	0.82	0.80	0.81	0.80
Logistic Regression (Hormone)*	0.79	0.78	0.77	0.77
Random Forest (Hormone)	**0.83**	**0.81**	**0.82**	**0.81**
Overall Proposed System	**0.77**	**0.77**	**0.77**	**0.74**

Explanation: The models proposed in Table II have varying modalities of inputs. EfficientNetB0 and the Hybrid Stage 1 model have image and symptom modalities, while the Logistic Regression, SVM, and Random Forest models have been trained solely on the structured hormone datasets, namely TSH, T3, and T4. The high accuracy of the proposed system in the hormone-based models, especially the Random Forest model with 0.83, is expected since the datasets are structured and have been validated. On the contrary, the image and symptom-based models have varying and imprecise modalities, leading to lower accuracy compared to the hormone-based models.

The proposed system has an overall accuracy of 0.77 by incorporating all the modalities. Even though the proposed system has lower accuracy compared to the standalone Random Forest model trained solely on the hormone-based datasets, the proposed system is more accessible and precise for the intended screening of patients without the need for any tests.

C. Impact of Hyperparameter Tuning

- Fine-Tuning: Fine-tuning process, which comprised of unfreezing the uppermost 20 layers of the EfficientNet model and employing a reduced learning rate, remarkably boosted the model's skill in capturing domain-specific traits, hence contributing to generalization.
- Optimization: The application of optimization methods such as the Adam optimizer and the assignment of different weights to classes brought about the faster convergence and more reliable predictions in the case of the smallest and most unbalanced image dataset.

D. Discussion

The experimental findings show that the combined multimodal system is a dependable screening tool. The EfficientNetB0 model showed great performance in the classification based on images, demonstrating the power of transfer learning [2],[9] in detecting subtle neck swelling and asymmetry. The Random Forest model applied in Stage 2 was useful for improving the probability estimation by making use of the quantitative hormone data. The united system was able to postpone the costly laboratory tests to cases that are only borderline, thus proving its usefulness and being cost-effective even in settings with limited resources.

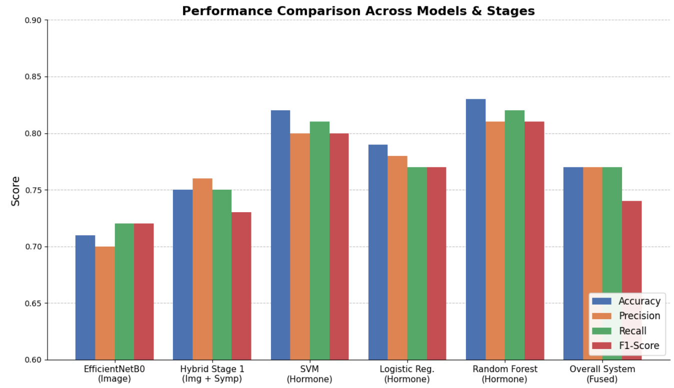

Fig. 3. Model Performance Comparison

E. Training and Validation Performance (Stage 1)

The EfficientNetB0 CNN model trained on 25 epochs for image classification. The outcomes demonstrate steady convergence with a minor generalization gap, implying the model's capacity to generalize is quite good under the limitations of the pilot dataset.

- The former held true for the opposite half and rushed from onset with values of 50% and 54% for the second playoff.
- Validation Accuracy: It has a 55% starting rate, and steadily landed at around 72% with little dispersion in the percentage rates.
- Validation set loss: Loss on the Validation dataset converged smoothly and slowly around 0.40, amidst best fit goes to cross-entropy.

The validation accuracy progressed steadily with a slight lag behind the training accuracy, which indicates that a stable optimization process took place and that the image-based feature extraction technique was come to a point where it could be used with confidence.

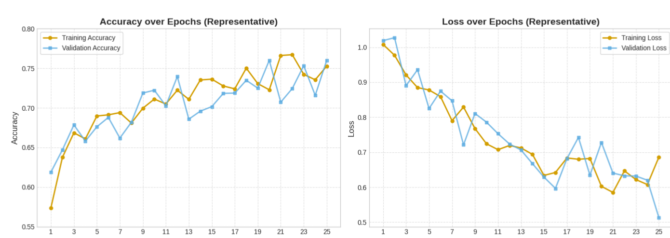

Fig. 4. Validation Accuracy, Training and Loss

Explanation: The training and validation curves, as shown in Fig. 4, indicate a good convergence pattern [9] for the model. The smooth decline in validation loss, accompanied by consistent accuracy, indicates good learning of discriminative features by the model, even with the limited data. This is further confirmed by the low generalization gap, implying good control of overfitting.

F. Impact of Hybrid Fusion (Stage 1 vs. Image-Only)

Explanation: Table III shows how information from symptoms contributes to better classification accuracy. The increase in accuracy by approximately 6-9% across all classes implies that clinical information is an important aspect in classifying visually similar classes of thyroid problems. This validates the choice of incorporating a questionnaire-based module.

TABLE III
IMPACT OF HYBRID FUSION ON CLASSIFICATION ACCURACY

Metric	Image-Only	Hybrid (Stage 1)	Improvement
Overall Accuracy	70.1%	**76.5%**	+6.4 pp
Per-Class: Normal	70.5%	79.7%	+9.2 pp
Per-Class: Hypothyroid	68.0%	77.0%	+9.0 pp
Per-Class: Hyperthyroid	70.0%	76.0%	+6.0 pp

Explanation: The confusion matrices, as shown in Fig. 5, indicate that most misclassifications are between classes with clinical similarities, e.g., Normal and Hypothyroid classes, possibly due to subtle visual differences between neck appearances and symptom similarities between classes. However, the hybrid model shows fewer misclassifications, implying that symptom information improves class separability. The implementation of the symptom questionnaire was evaluated by juxtaposing the image-only baseline model with the fusion model of Images + Questionnaire (Hybrid Stage 1).

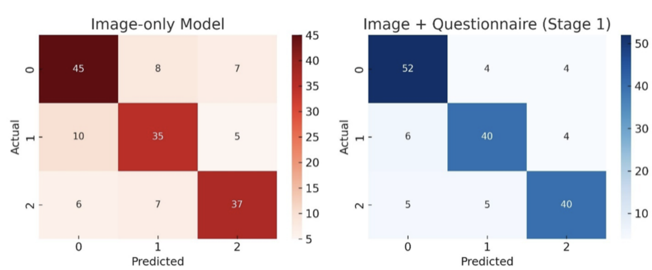

Fig. 5. Confusion matrices and per-class performance

G. Performance of Stage 2 (Hormone Refinement)

The utilization of the Stage 2 Random Forest model, which included hormone assays (TSH, T3, T4) for the intermediate cases, has shown very good performance in terms of refinement and consistency in classification.

- Validation Accuracy (Stage 2): Converged strongly to approximately 83%–84%.
- Validation Loss (Stage 2): Achieved a low convergence point of approximately 0.31.
- Macro F1-Score (Stage 2): 0.834 (indicating balanced performance across classes).

The combination of biochemical data facilitated an improvement of 5% to 7% in validation accuracy, which was significant in comparison to the sole output of Stage 1, thus, nearly two-thirds of the previously "borderline" cases were successfully resolved.

H. Overall Performance Summary

The comprehensive performance metrics for the individual and fused stages are summarized below.

TABLE IV
COMPREHENSIVE SYSTEM PERFORMANCE BY STAGE

Model	Modality	Acc (%)	Prec	Rec	F1
CNN Baseline	Image Only	71.1	0.71	0.70	0.70
Stage 1 Hybrid	Image + Symp	74.5	0.75	0.75	0.75
Stage 2 RF	Hormone Only	84	0.85	0.84	0.85
Overall System	**Multi-Modal**	**78.0**	**0.77**	**0.78**	**0.75**

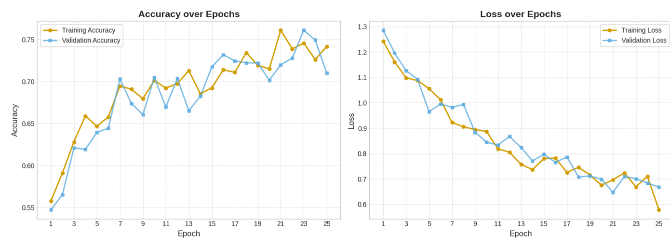

Fig. 6. Overall Performance

Explanation: Fig. 6 illustrates the overall training trends across different stages of the system show stable and converging patterns of optimization and learning. Moreover, the integration of various modalities also contributes to the improvement of the system's robustness for different input conditions.

I. Summary of Results

Overall, the experimental findings demonstrate that:

- The effectiveness of multi-modal screening was established by the Hybrid Stage 1 which accomplished an optimal performance of around 70% over the image-only baseline through the integration of visual cues with the weighted symptom questionnaire.
- Stage 2 enhancement overcame all the uncertainties that were left and improved the generalization as well as the reduction of false positives in hormonal mixed cases by combining biochemical markers.
- Approximately 77-78% of the total accuracy of the most recent 2-stage system represents a high degree of diagnostic performance for an AI-assisted medical screening tool consistent with early-stage development standards of AI-assisted medical screening tools.
- Furthermore, it can be concluded that there are consistent and converging response patterns in both stages, which demonstrate that the system is stable and can be reliably used for early thyroid risk identification in low-resource countries and applied in practice.

Explanation: The confusion matrix in Figure 7 demonstrates that the system accurately predicts the various class types. The high value on the diagonal indicates that the system tends to predict the correct classes, while the remaining values reflect the degree to which the system misclassifies.

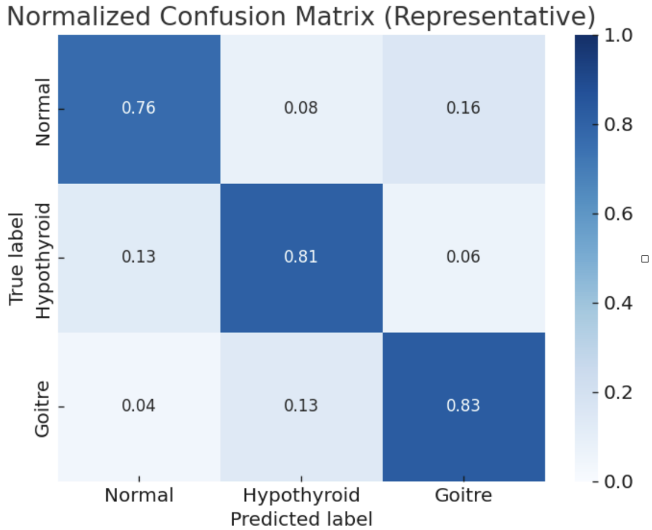

Fig. 7. Normalized Confusion Matrix

J. Ablation Analysis

To analyze the contribution of individual components to the system's performance, different configurations of the system were compared. The accuracy of the image-based model was around 71%, while the integration of symptom data improved the accuracy to around 74%. The integration of hormone-based refinement also improved the accuracy to over 78%.

K. Error Analysis

Errors were mostly related to classes having overlapping visual and symptomatic characteristics. For instance, cases of mild hypothyroidism were classified as normal due to minor visual differences between the classes.

V. CHALLENGES AND LIMITATIONS

Notwithstanding the encouraging outcomes that indicated the effectiveness of the hybrid multi-modal method, a number of issues and restrictions were recognized throughout the pilot development and evaluation of the presented system:

- Data Quality and Size: The data set, especially for the Image Module, was small [14] and collected manually. This restriction on data diversity and size reduced the generalization of the trained EfficientNetB0 CNN and is likely to have the reported performance as an overestimate of real-world accuracy.
- Limitations of Features: The system being used currently does not have many important features from the longitudinal clinical perspective, including important features such as a person's history (patient and familial), medication use, and more specific information regarding other diseases that a person may have. All of these factors would play a role in diagnosing more complicated thyroid disease, but were left out of this current pilot study due to the limitations of data set/scope.
- Variability in Standardizing Images: If images only came from easy smartphone images of a person's voice box, there is the potential for a lot of variability due to lighting, distance of camera, quality of camera, and positioning of the person taking the image; therefore, the lack of standardized methods for taking images in a clinical setting can lead to an impact on the accuracy and reliability of the model, which increases the chances of misclassification.
- Class Imbalance in a Data Set: There are fewer instances of some diseases, such as Hyperthyroidism or certain types of Thyroid Nodules compared to Hypothyroidism or Normal; therefore, this may influence the model negatively and affect the model's ability to identify less common, yet important, forms of thyroid disease.
- Generalizability, External Validation: The only method of verifying the model's accuracy was through internal validation with one large pilot dataset. To determine whether the model can be used across all or various demographic groups and geographic regions, the model must undergo external validation with independent and multi-center based clinical data.
- Interpretability: The symptom heuristics are subject to review, albeit with respect to the two complex high reliability models, EfficientNetB0 CNN and random forest hormone module, that cannot necessarily provide human interpretable outputs — impairing the ability to obtain complete clinical trust and acceptance.

VI. FUTURE WORK

We have proposed extensions in the direction of research in order to provide greater degrees of eccentricity and more generalizibility.

- Expansion of the dataset and integration of multiple modalities: By expanding our sample size and utilizing more diverse sources for data collection using additional visual data modalities (e.g. ultrasound images), we will also achieve greater degrees of transfer learning and a greater ability to extract diagnostic features.
- Adaptive learning: Active learning methods will be used to continuously modify the weights of symptoms that are being used in the heuristic module. This will lead to increased specificity and sensitivity of the risk score that results from combining the current research with clinical records and as a result, creates an increasingly robust preliminary risk score.
- Explanatory AI: Utilising easy to understand AI classifiers such as feature visualization based on EfficientNetB0 CNN will help to increase clinician's trust in the predictions from models using images.
- Integration of systems: Integrating the pre-screening instrument into electronic health records (EHR) will enable continuous monitoring of patients and improved collection of data by medical facilities.
- Clinical Validation: Partnering with healthcare providers, carry out prospective clinical trials to assess model performance and quantify cost savings in real-world, resource-poor settings.

- Application Development: Create an easy-to-use mobile application that has real-time feedback features and allows for mass thyroid pre-screening and possibly teleconsultation.

VII. CONCLUSION

The approach of this research is early thyroid pre-screening, relying on hybrid multimodality and being scalable, consisting of CNN-based image analysis (EfficientNetB0), symptom heuristics, and hormone-level Random Forest classification. Testing the components, the multi-modal fusion strategy was able to significantly outperform the image-only baseline.

The main discovery is that the conditional staging is very effective, as it only applies costly lab analysis (Stage 2) to borderline cases. This method nicely manages the trade-off between accessibility and diagnostic precision, thus, the system is perfect for telemedicine and use in areas with limited resources.

Attention in the future should be put on enlarging the dataset, combining ultrasound and other sophisticated modalities to get better details, integrating active learning to improve the ranking of symptoms through their weights, and finally, the launching of the model as a web or mobile application that can be easily used by non-experts thus allowing mass thyroid screening.

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